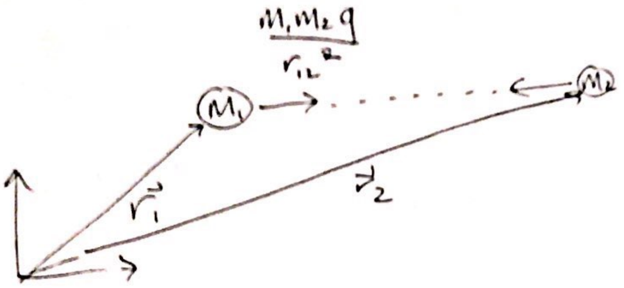
MAE 4730
9/10/18
Lecture 7

Today: multi particle systems

For one particle:

$\vec{F} = m\vec{a}$ — For many particles $*$ applies to each particle
$\vec{F}$ includes forces from other particles

ex) 2 particles w/ gravitational attraction



$m_1\vec{a}_1 = \vec{F}_{12}$

$m_2\vec{a}_2 = \vec{F}_{21}$

$$\vec{F}_{12} = -\vec{F}_{21} = \frac{(\vec{r}_2 - \vec{r}_1)\, m_1 m_2 G}{|\vec{r}_2 - \vec{r}_1|^3}$$

Initial conditions
$\vec{r}_1(0)$, $\vec{r}_2(0)$
$\vec{v}_1(0)$, $\vec{v}_2(0)$

In 3D — how many 1st order ODE's

$12 \xrightarrow{\text{reduces}} 4$
- solve 8 in trivial ways
- system as whole no force
- m_{cg} no force

$$\underbrace{z = \left.\begin{bmatrix} [\vec{r}_1] \\ [\vec{r}_2] \\ [\vec{v}_1] \\ [\vec{v}_2] \end{bmatrix}\right\} 12}_{\text{organize like this}}$$

Matlab:
r=[r1;r2] ; v=[v1;v2]
rdot=v ; % 6 ODE's

ex) Any collection of n particles w/ all forces of form $\vec{F}_i = \vec{f}(\vec{r}_1, \vec{r}_2, \ldots, \vec{v}_1, \vec{v}_2, \ldots, t)$ — total force on particle i

For each particle

$\vec{F}_i^{net} = \vec{F}_i^{net}(\vec{r}_j, \vec{v}_j, t) = m_i\vec{a}_i$ — n equations of this form

$j = 1, n$

$\Rightarrow$ 6n ODES (first order)

$\Rightarrow$ Any system modeled as a collection of particles w/ force laws we know, we can solve for its motion

Force Laws:

ex) $\vec{F} = -c\vec{v}$

$\vec{F} = -c|\vec{v}|\vec{v}$] from environment

ex) $\vec{F}_{12} = c(\vec{v}_2 - \vec{v}_1)$] dashpot connection

on particle 1 from particle 2

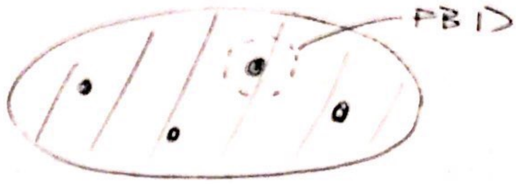
ex) $\vec{F}_{12} = k\left(|\vec{r}_2 - \vec{r}_1| - l_0\right)\underbrace{\dfrac{\vec{r}_2 - \vec{r}_1}{|\vec{r}_2 - \vec{r}_1|}}$

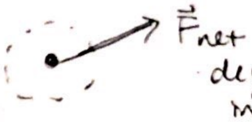
"kx" spring connection

ex) Gravity

ex) Electrostatics

ex) $\underbrace{\text{lead balls}}_{\text{mass}}$ in $\underbrace{\text{styrofoam}}_{\text{structure}}$

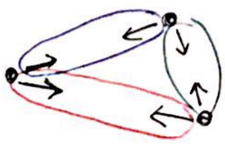




depends on all the atoms in balls

Theorems

- Usual main assumption:
  - Forces are pairwise equal + opposite



|| BAD ||

1) None of our business
   - Makes mechanics depend on microscopic physics that we don't know
2) Wrong
3) Makes wrong macroscopic predictions
   - predictions about elastic moduli that are wrong

⇒ We want better assumptions

(can read about in Wilchek (Physics today)

Truesdel ("whence the moment of momentum")

Axioms of multi-particle dynamics:

I. $\vec{F} = m\vec{a}$ for each particle

II. $\vec{F} = \vec{F}_{int} + \vec{F}_{ext}$

forces from outside system

forces from inside system

IV. Several Equivalent choices (for any system or subsystem)

A. Principals of LMB + AMB

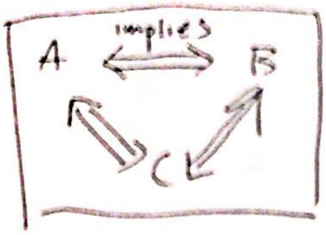
LMB: $\sum \vec{F}_j^{ext} = \sum m_i \vec{a}_i$

AMB: $\sum \vec{r}_{j/c} \times \vec{F}_j = \sum \vec{r}_{i/c} \times m_i \vec{a}_i$ for all points C

B. All internal forces are in self equilibrium

$$\underset{\text{all particles + forces}}{\sum\sum} \vec{F}_i = \vec{0} \quad \Big) \text{ no net force} \qquad \sum\sum \vec{r}_{i/k} \times \vec{F}_i = \vec{0} \quad \Big) \text{ no net moment}$$

C. All internal forces do no work for rigid moments of the particle assembly



All force laws you will use are consistent w/ this

⇒ when we solve the Gen ODEs, the solutions always obey the theorems

Simplified ways to think about systems of particles:

LMB: $\sum \vec{F}^{ext} = \sum m_i \vec{a}_i$

$$= \frac{d}{dt} \sum m_i \vec{v}_i$$

$$= \frac{d^2}{dt^2} \sum m_i \vec{r}_i$$

$$= \frac{d^2}{dt^2} \left(M_{tot} \vec{r}_{G} \right)$$

define $\cancel{\vec{r}_G}$ $\boxed{M_{tot} \vec{r}_G = \sum m_i \vec{r}_i}$

For any system,

$$\boxed{\vec{F}^{net} = M_{tot} \vec{a}_G}$$

ex)